Vedant Desai

(248) 704 – 4852 | vedantde@umich.edu | github.com/Verdent06 | linkedin.com/in/vedantde06

EDUCATION

University of Michigan — Expected May 2028
B.S. in Computer Science and Economics — Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Discrete Math, Calc III

EXPERIENCE

Vylet — May 2026 – Present | vyletdata.com | GitHub
Founder — Automated lead-sourcing platform for PE/search-fund acquisition prospecting — live product generating $1,500 MRR across three paying clients

- Launched Vylet, automating a 30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Engineered a LangSmith eval pipeline spanning 20 adversarial business test cases across 13 archetype labels — manufacturers, SaaS tools, PE holding companies, geographic mismatches — then layered deterministic Pydantic consensus gates to lift extraction faithfulness from 50% to 90%.
- Grew Vylet from 1 to 3 paying subscription clients within six weeks of launch — spanning workforce-software, landscaping, and geography-first sourcing engagements — generating $1,500 in monthly recurring revenue.

CaseStudyPrep.AI — Dec 2025 – May 2026
Software Engineer Co-op (Voice AI) — Remote

- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.
- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.

Michigan Data Consulting (MDC) — Jan 2026 – May 2026
Data Engineer — Michigan Campaign Finance Network — Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at 2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating 800 hours of manual pulls across 400 tracked PACs.
- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.

PROJECTS

SignalWeaver | *PyTorch, Transformers, LoRA, scikit-learn, pgvector* — github.com/Verdent06/SignalWeaver
Multi-signal financial research platform combining fundamentals, macro indicators, and news sentiment (research assistant; not investment advice)

- Lifted financial-sentiment classification accuracy from 81% to 96% by LoRA fine-tuning a quantized meta-llama/Meta-Llama-3.1-8B-Instruct model on 3,454 Financial PhraseBank entries, evaluated on a held-out test set.
- Implemented semantic search over stored financial news with pgvector cosine similarity (768-d MPNet embeddings), hitting 49ms p50 / 99ms p99 over 90 queries returning top-5 or top-10 matches.
- Built a linear regression combining fundamental, embedding-based sentiment, and LLM-derived sentiment signals into a composite return-prediction score; validated out-of-sample (3.39% R^2, consistent with the low single-digit R^2 typical of real return-prediction literature) to confirm the model wasn't just fitting noise.

TECHNICAL SKILLS

AI / ML .... PyTorch, scikit-learn, LangGraph, LangSmith, Silero VAD, ONNX Runtime, Pandas
Languages .... Python, C++
Frameworks .... Flask, Angular, RxJS
Databases .... pgvector
Cloud & Infrastructure .... AWS (EC2, S3), Docker, Redis, Celery, Git